Warehousing & Contract Logistics Reference

How goods are stored, handled, and managed between transport legs — facility types, regulatory categories, and the value-added services that make up contract logistics

1. What Warehousing and Contract Logistics Cover

Warehousing is the storage of goods between two points in a supply chain, whether that gap is a few hours between unloading a container and loading it onto an outbound truck, or several months of seasonal stock-building ahead of a peak sales period. Contract logistics extends beyond simple storage to cover the full set of activities a business would otherwise have to run itself: receiving, put-away, inventory management, order picking, packing, labelling, and onward dispatch, typically carried out under a multi-year agreement rather than a one-off booking. The distinction matters commercially, because a warehousing contract is usually priced on space and duration, while a contract logistics agreement is priced on throughput, service levels, and the specific activities performed on the goods while they are in custody.

A business chooses to outsource warehousing and contract logistics for reasons that go beyond simply not wanting to run a warehouse. Building and operating a facility ties up capital in land, racking, handling equipment, and a trained workforce, all of which are difficult to scale up or down as demand shifts through the year. A logistics provider that already operates multiple facilities can absorb a client's seasonal peak within existing capacity, share specialised equipment such as reach trucks or temperature-controlled chambers across several clients, and offer a service level agreement backed by trained staff and an existing warehouse management system, rather than requiring the client to build all of that from scratch.

The commercial relationship between warehousing and contract logistics is also worth understanding as a spectrum rather than two entirely separate offerings. At the simplest end sits pure space rental, where a client leases square footage or pallet positions and handles all the receiving, picking, and dispatch work with its own staff. At the other end sits a fully managed contract logistics operation, where the provider's own staff, systems, and equipment perform every physical touch on the goods, and the client interacts with the operation almost entirely through data — stock levels, order status, and service-level reporting — rather than through any physical presence at the facility. Most real arrangements sit somewhere between these two ends, with the specific mix of self-managed and provider-managed activity negotiated according to what the client's own organisation is best placed to do internally.

2. Types of Warehousing Facilities

Not every facility that stores goods performs the same function, and the terminology used across the industry reflects real differences in location, regulatory status, and the kind of cargo each is built to handle.

Choosing the right facility type for a given shipment is less about which option is generically “better” and more about matching the facility's function to the specific stage a shipment is at and the nature of the cargo itself. A shipment still awaiting customs clearance has no practical use for a general distribution warehouse, since that facility type is not licensed or equipped to hold goods under bond or manage a customs examination. Equally, cargo that has already cleared customs and simply needs to reach a retail shelf quickly gains little from sitting in a CFS, whose systems and staffing are built around customs-adjacent activity rather than fast retail throughput. Understanding which facility type a shipment is currently sitting in is therefore one of the fastest ways to understand what is actually happening to it at that moment — whether it is awaiting a regulatory process, awaiting onward transport arrangement, or simply moving through routine distribution.

General and Distribution Warehouses

These are the most common facility type: open-market storage used for domestic goods that have already cleared customs, or that never needed to. A distribution warehouse is usually positioned close to a consumption centre or a major road and rail corridor, and is built around fast inbound and outbound movement rather than long-term storage, since the goods passing through are typically destined for retail shelves, e-commerce fulfilment, or onward manufacturing within days or weeks.

Container Freight Stations and Inland Container Depots

A Container Freight Station, commonly abbreviated CFS, is a facility located near a port where containers are stuffed, destuffed, and consolidated, and where customs examination of cargo frequently takes place before release. An Inland Container Depot, or ICD, performs a similar function but is located away from the coast, closer to inland manufacturing or consumption hubs, and is connected to the port by rail or road so that containers can be customs-cleared without the cargo ever having to travel to the port itself. Both facility types sit at the intersection of warehousing and customs administration, and a shipment's dwell time is frequently measured from the moment it enters one of these facilities.

Cold Storage and Temperature-Controlled Facilities

Perishable cargo — pharmaceuticals, fresh produce, dairy, and certain chemicals — requires storage within a defined temperature band, monitored continuously rather than checked periodically. These facilities carry higher construction and running costs than ambient warehousing because of the refrigeration plant, insulated construction, and backup power required to maintain the cold chain during a grid outage, and they are typically priced accordingly.

3. Bonded Warehousing versus Open Warehousing

The single most consequential distinction in warehousing for an import or export business is whether a facility is bonded or open, because this determines when import duty falls due and how much flexibility the cargo owner has while goods are in storage.

Bonded Warehousing

A bonded warehouse is a facility licensed by the customs authority to store imported goods before duty has been paid. Goods entering a bonded facility are placed under a customs bond, meaning duty and applicable taxes are deferred until the goods are actually withdrawn for domestic consumption, rather than becoming payable the moment the goods land in the country. This is genuinely useful for a business that imports in bulk but sells gradually: duty is paid only against the portion of stock actually released to the market, which improves cash flow considerably compared with paying duty on the full consignment up front. Bonded storage is equally common for re-export scenarios, where goods pass through a country without ever entering the domestic market and duty is therefore never triggered at all. The trade-off is administrative: bonded facilities operate under closer customs oversight, movement of goods in or out must be documented and, in many cases, authorised in advance, and there is typically a maximum period goods may remain under bond before duty becomes payable regardless of whether the goods have been sold.

Open Warehousing

An open, or duty-paid, warehouse stores goods that have already cleared customs and on which duty has been settled. Once goods reach this stage they are treated the same as any domestically produced stock: they can be moved, repacked, sold, or transferred between facilities without further customs involvement. Most day-to-day distribution and retail fulfilment activity happens out of open warehousing, since the customs process has already concluded by the time goods reach this stage of the supply chain.

A business handling regular import volumes will often use both facility types in sequence: goods land and sit in a bonded facility while duty liability is assessed and deferred, and once a batch is released against a confirmed sale or a scheduled distribution run, that batch moves into open warehousing for onward handling. Structuring storage this way keeps duty payments aligned with actual cash inflow rather than with the date the goods happened to arrive in the country.

The maximum period goods may remain under bond varies by jurisdiction and, in some cases, by commodity category, but it is never indefinite. Once that maximum bonded period is reached, customs authorities typically require duty to be paid regardless of whether the goods have been sold or are still awaiting a buyer, and goods left unclaimed beyond an even longer outer limit can, in some jurisdictions, be subject to disposal or auction by the customs authority itself. A business planning to use bonded storage as a deliberate cash-flow strategy should therefore track the bonded clock for each consignment as closely as it tracks the underlying sales forecast, since a shipment that sits under bond longer than planned converts an intentional deferral into an unplanned duty liability with very little additional warning before it comes due.

4. Dwell Time — What It Is and Why It Matters

Dwell time is the length of time cargo remains within a facility — a port terminal, a CFS or ICD, or a warehouse — between one movement event and the next, most commonly measured from the moment a container is discharged from a vessel to the moment it is gated out for onward delivery. It is one of the most closely watched metrics in contract logistics because it drives cost on both sides of the relationship: the facility operator, because storage space occupied by slow-moving cargo cannot be used for new inbound volume, and the cargo owner, because most facilities charge escalating storage fees the longer a container or consignment remains uncollected.

Dwell time is driven by a mix of factors that are within a shipper's control and factors that are not. Within the shipper's control: how quickly import documentation is filed and duty is paid, whether the consignee has arranged onward transport in advance, and whether the paperwork accompanying the shipment is complete and consistent on first submission. Outside the shipper's control: how quickly customs completes an examination if the shipment is selected for one, congestion at the facility itself during a demand surge, and delays caused by other regulatory bodies, such as a plant quarantine or drug-testing authority, that must sign off on certain categories of cargo before release.

Extended dwell time has consequences beyond the storage fees themselves. Facilities running near capacity because of slow-moving cargo experience knock-on congestion that increases dwell time for other, unrelated shipments passing through the same yard, and container lines that see their equipment tied up for longer than planned may apply detention charges on top of the facility's own storage fees. For this reason, most contract logistics agreements build dwell time targets directly into the service level agreement, with the provider responsible for flagging documentation gaps early enough that they do not become a release delay.

Most facilities price storage on an escalating scale specifically to discourage long dwell times: an initial free period covers the time reasonably needed to complete customs formalities and arrange onward transport, after which a daily rate applies, and that daily rate frequently increases again after a second, longer threshold. This structure reflects the facility's own economics — a container sitting well past a reasonable clearance window is occupying space that could otherwise turn over several times for other customers — and it is also why a shipment that seems to have cleared customs without issue can still generate a meaningful storage bill if the consignee is slow to arrange final pickup. A customer surprised by a storage charge on an otherwise problem-free shipment has often simply exceeded the free period on the collection side rather than experienced any delay in the clearance process itself.

5. Third-Party Logistics (3PL) Operations

A third-party logistics provider, generally shortened to 3PL, takes over one or more physical logistics functions on a client's behalf — warehousing, transportation, or both — under an ongoing commercial arrangement rather than a single transaction. The defining feature of a 3PL relationship is that the provider takes physical custody and operational control of the client's goods, which distinguishes it from a freight forwarder or customs broker, whose role is arranging and documenting movement rather than physically handling and storing the goods.

Core 3PL Functions

bullet Inbound receiving and quality inspection — verifying that a delivery matches the purchase order or advance shipping notice before it is accepted into inventory.

bullet Put-away and inventory management — placing goods in the correct storage location and maintaining an accurate, real-time record of stock on hand.

bullet Order picking and packing — assembling individual customer or store orders from bulk inventory, and packing them to the client's specification.

bullet Outbound dispatch and last-mile coordination — consolidating outbound loads and handing them to the appropriate transport mode, whether that is a full truckload carrier, a parcel network, or the client's own fleet.

bullet Returns processing — receiving returned goods back into the facility, inspecting condition, and routing them to resale, refurbishment, or disposal as appropriate.

The scope of a 3PL contract is negotiated activity by activity rather than assumed. A client running a lean e-commerce operation may want the provider to handle everything from receiving through to last-mile handover, while a manufacturer with its own distribution fleet may only want warehousing and outbound loading, keeping transport under its own control. Because pricing is usually tied to the specific activities performed and the volume passing through the facility, a clearly scoped contract at the outset avoids disputes later over what is and is not included in the agreed rate.

It is also worth distinguishing a 3PL relationship from a fourth-party logistics, or 4PL, arrangement, since the two terms are sometimes used loosely. A 3PL provider physically executes the logistics activity itself — it is the entity whose warehouse the goods sit in and whose staff pick and pack the orders. A 4PL provider, by contrast, typically does not operate its own physical facilities at all; instead it manages and coordinates a network of underlying 3PL and transport providers on the client's behalf, acting as a single point of accountability across a multi-provider supply chain. A client with operations spanning several countries or several different 3PL partners may specifically want a 4PL layer on top to provide a single, consolidated view across all of them, whereas a client with a single facility and a single provider generally has no need for that additional layer.

6. Value-Added Services in Contract Logistics

Beyond storage and movement, many clients ask their logistics provider to perform light processing on goods while they are already in the facility, since doing so avoids an extra handling step and an extra transport leg elsewhere in the supply chain. These are generally described as value-added services, and the range on offer varies considerably between providers and facility types.

bullet Kitting and assembly — combining several individual components into a single saleable unit, such as bundling an accessory with a primary product ahead of dispatch.

bullet Labelling and re-labelling — applying country-specific compliance labels, barcodes, or retailer-mandated tags before goods leave the facility.

bullet Quality control and inspection — checking a sample or the full batch of an inbound consignment against agreed specifications before it is accepted into saleable inventory.

bullet Repackaging — breaking down bulk shipping packaging into retail-ready units, or the reverse, consolidating retail units into bulk cartons for a specific customer.

bullet Reverse logistics and refurbishment — managing the return, inspection, and where appropriate the refurbishment or repackaging of goods sent back by end customers.

Value-added services are typically priced separately from base storage and handling, since the labour and equipment involved vary widely by activity. A client evaluating a contract logistics proposal should look closely at how these services are costed individually rather than assuming they are bundled into a single all-in storage rate.

7. Warehouse Management Systems and Technology

A warehouse management system, or WMS, is the software layer that tracks every unit of inventory from the moment it is received to the moment it leaves the facility, and directs warehouse staff on where to place, pick, and move stock. In a facility of any meaningful size, a WMS is what makes it possible to know, at any given moment, exactly how much of a given item is on hand, where it is physically located, and which orders it has already been allocated against — information that would be effectively impossible to maintain reliably on paper or in a spreadsheet once volumes reach even a moderate scale.

Barcode and radio-frequency scanning are the most common way a WMS captures movement in real time: a worker scans a location and an item together at every put-away, pick, and dispatch step, and the system updates its inventory record instantly rather than relying on a periodic manual count. Larger or more automated facilities extend this further with fixed scanning tunnels, conveyor-integrated weight and dimension checks, and in some cases automated storage and retrieval systems that remove manual handling from repetitive, high-volume picking tasks altogether.

For a client, visibility into the WMS — typically through a client-facing portal or a regular data feed — is usually the single most valued piece of technology in a contract logistics relationship, since it lets the client's own planning and customer service teams answer stock and order status questions without calling the warehouse directly. Integration between the provider's WMS and the client's own order management system is increasingly treated as a baseline expectation rather than a premium add-on.

The maturity of a facility's WMS also has a direct bearing on how accurately dwell time and stock accuracy can be reported back to a client. A facility relying on periodic manual counts can only ever report inventory as of the last count, with real accuracy degrading the longer the interval since that count. A facility with full barcode or RFID scanning at every movement step can instead report a stock position that is accurate to within minutes of the actual physical state of the warehouse, which matters considerably for a client trying to make same-day fulfilment or replenishment decisions. When evaluating a prospective provider, the specific scanning technology in use at the point of receiving, put-away, and picking is usually a better indicator of true system accuracy than the WMS software brand name alone, since two facilities running the same software can still differ substantially in how consistently their staff actually use it at every touchpoint.

8. Safety, Compliance, and Quality Standards

A warehouse handling third-party goods carries obligations that go beyond simply keeping the building secure. Fire safety systems, appropriate racking design and load limits, trained forklift and reach-truck operators, and clearly marked pedestrian and vehicle routes within the facility are baseline requirements, and larger facilities are typically certified against recognised safety and quality management standards that are audited on a recurring basis rather than assessed once at commissioning.

Facilities handling regulated categories of goods — pharmaceuticals, hazardous chemicals, or bonded cargo under customs supervision — operate under additional layers of control specific to that category, such as segregated storage areas, restricted access logs, and documented handling procedures that a general-purpose warehouse would not need to maintain. A client evaluating a facility for regulated cargo should confirm the specific certifications and licences the facility holds for that category, rather than assuming general warehousing competence extends automatically to specialised handling requirements.

Insurance coverage is a related but separate matter from safety certification. A contract logistics agreement should specify whether the provider's insurance covers the full declared value of goods in storage, the claims process in the event of loss or damage, and whether the client is expected to carry supplementary cargo insurance of its own.

It is worth being explicit that safety certification and insurance coverage answer two different questions and neither substitutes for the other. A facility's safety certification speaks to how well the operator manages physical risk — the likelihood that a fire, structural failure, or handling accident occurs in the first place. Insurance coverage speaks to what happens financially if a loss occurs despite those precautions, and it is entirely possible for a well-certified, well-run facility to still carry insurance limits that fall short of a client's actual stock value, particularly during a seasonal peak when inventory on hand is unusually high. A client should review both separately, and should specifically confirm whether the facility's standard insurance policy caps the payout per incident, per client, or per square metre of storage, since any of these caps could leave a genuine gap relative to the client's own declared stock value at any given point in time.

Taken together, the facility types, regulatory categories, and service models covered in this reference describe a continuum a shipment typically moves along rather than a set of unrelated options: cargo frequently starts in a CFS or ICD for customs processing, may pass through bonded storage while duty liability is deferred, and ultimately reaches open warehousing or a distribution centre for final fulfilment, with a 3PL provider potentially managing several of these stages under a single contract logistics agreement. Recognising where in this continuum a specific shipment currently sits is usually the fastest way to understand both what has already happened to it and what remains before it reaches its final destination.

This guide is a summary reference for warehousing and contract logistics practice. Specific facility certifications, bonded storage periods, and service level terms should be confirmed directly with the relevant facility or provider for any particular arrangement.